MINERvA OmniFold

Research catch-up with Ben

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2D validation, higher-dimensional inference, and full-event learning

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The project now has three distinct outcomes

2D reproduction	Validated central result and a rebuilt uncertainty construction
3D to 5D scalar studies	Validated finite closure tests and conditional generator inference
Full-event PET studies	Better recovery in simulation, but no eligible uncertainty procedure

The data and the measurement

Sample	Medium-energy neutrino mode, about 6 GeV. 4.12 million selected events
Signal	Muon-neutrino charged-current interactions in the tracker
Reference domain	Muon angle $< 20°$, $1.5 < p_{\parallel} < 60$ GeV/c, $pT < 4.5$ GeV/c
Detector	MINERvA measures activity. Downstream MINOS measures the muon

OmniFold learns event weights in two spaces

01

Detector space

Compare data with reconstructed simulation

Fit a weight for each simulated event

02

Paired simulation

Carry that weight to its truth record

Estimate weights for missed events

03

Truth space

Learn a smooth truth-level reweighting

Repeat with the updated simulation

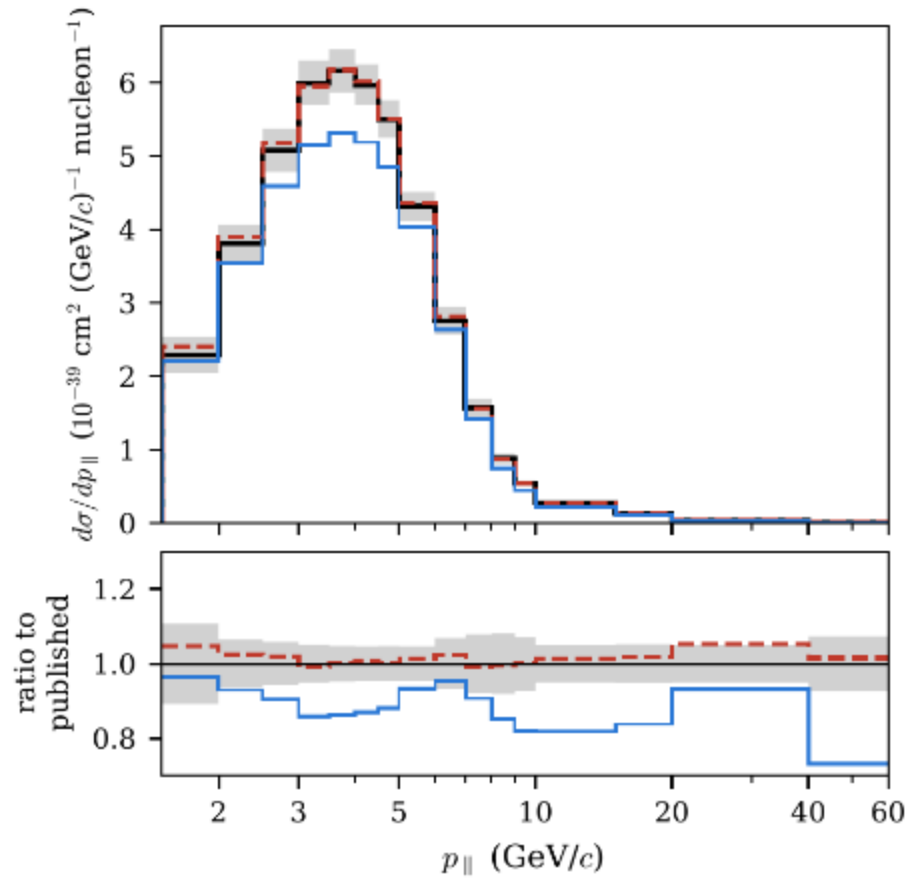
After five iterations, histogram the weighted truth events in the desired observables

Three different tests of an unfolding result

Closure	Create simulated data with known truth and test recovery
Dimensional anchor	Integrate an added observable and compare with the earlier result
Coverage	Repeat simulated experiments and count how often intervals contain truth

A closure pass applies to the tested deformation and detector response

The 2D cross section reproduces the published scale



Black: published Red dashed: OmniFold Blue: tuned prior

3.073 × 10^{-38} cm²/nucleon

1.11% above the published total

205 reported bins. Five iterations

Same data and shared systematics

Projection of the validated 2D central result. This is a reproduction, not an independent measurement

The 2D controls test recovery and bookkeeping

Injected shapes	Truth reweights and alternative-universe tests pass their declared thresholds
Missed events	Truth records define the full signal denominator
Iteration controls	Five iterations form the frozen production choice
Known limit	A large hidden response deformation exceeds the closure threshold

The rebuilt uncertainty recipe still needs a coverage test

Construction	187 systematic universes + statistics + training variation
Current budget	6.87% median relative uncertainty per reported bin
Old statistical band	Two-standard-deviation containment: 91.21%. Failed its window
Repair	Adopted VL170 band, rebuilt in VL172. Coverage not re-tested

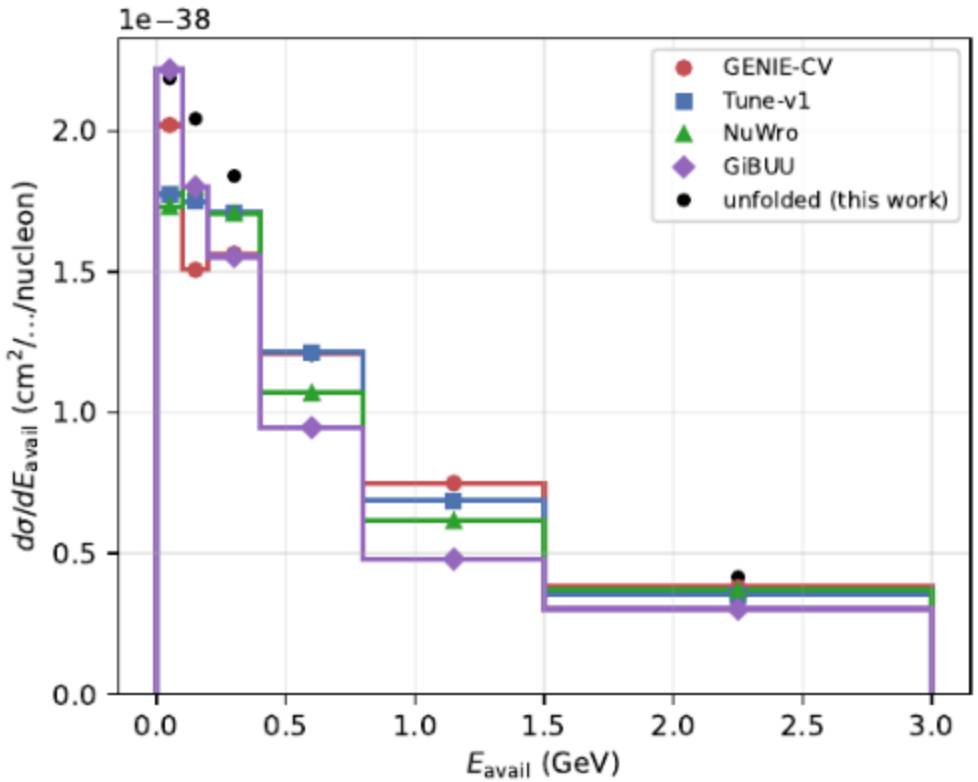
The 6.87% is a median bin uncertainty, not the uncertainty on the integrated cross section

Each added observable asks a physics question

2D: pT, pl	Muon kinematics anchor the published measurement
3D: add Eavail	Visible hadronic energy resolves recoil structure
4D: add q3	Magnitude of momentum transfer adds interaction information
5D: add W	An inferred hadronic mass separates regions of the spectrum

“5D” counts recorded observables. It does not imply five independent degrees of freedom

Hadronic energy reveals model differences



3D normalization anchors to 2D

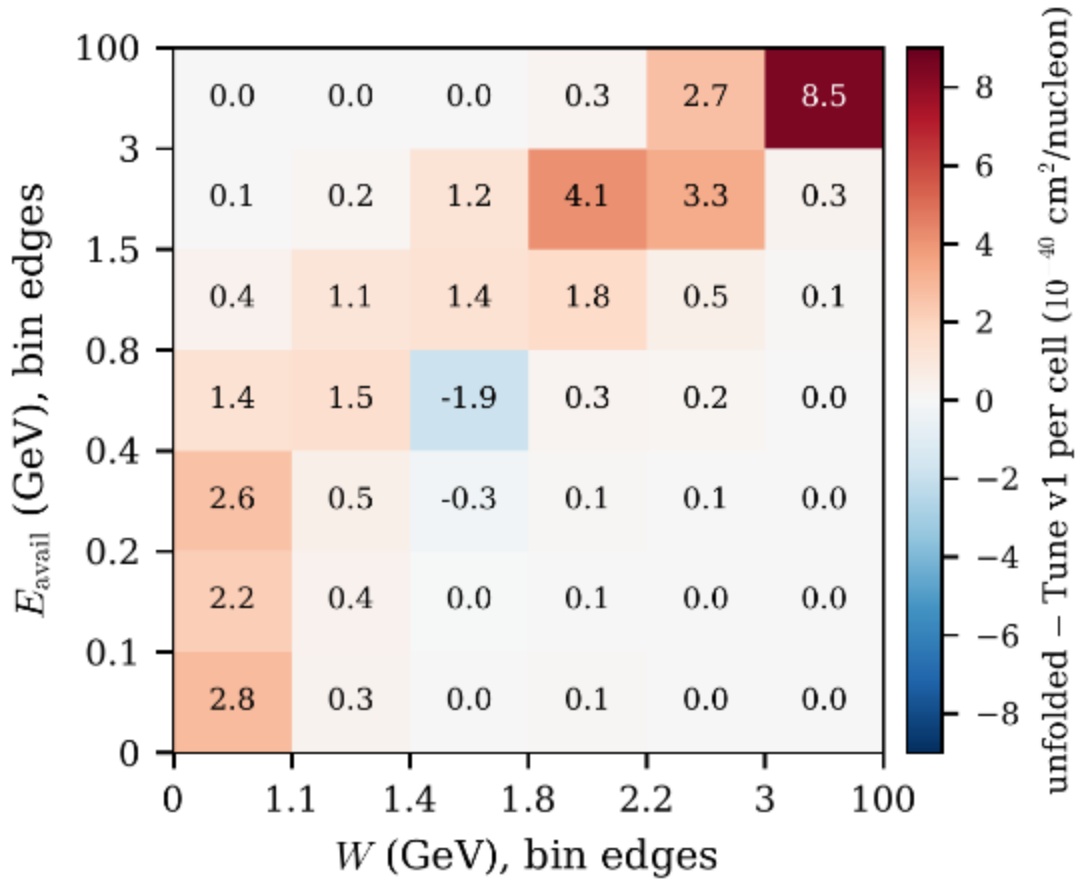
The tested Eavail injection closes

Valencia 2p2h fills 52% of the gap at Eavail ≤ 0.4 GeV

Central-value comparison only

Generator predictions use repaired samples. Historical covariance significances stay unquoted

The 5D projection locates a high-energy difference



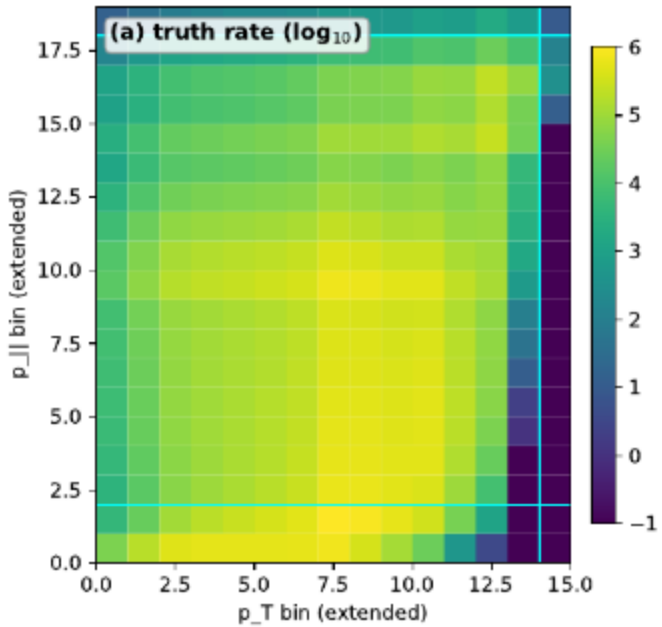
Added-axis anchors pass

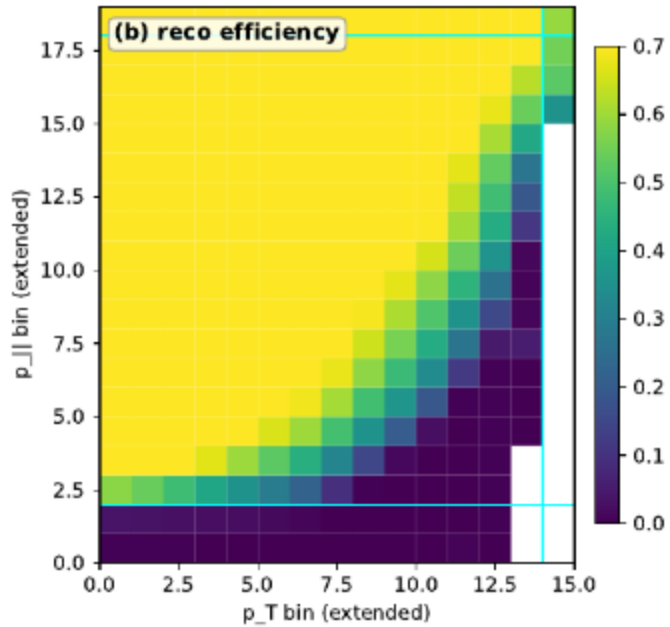
The nominal result exceeds the tuned prior at high Eavail and W

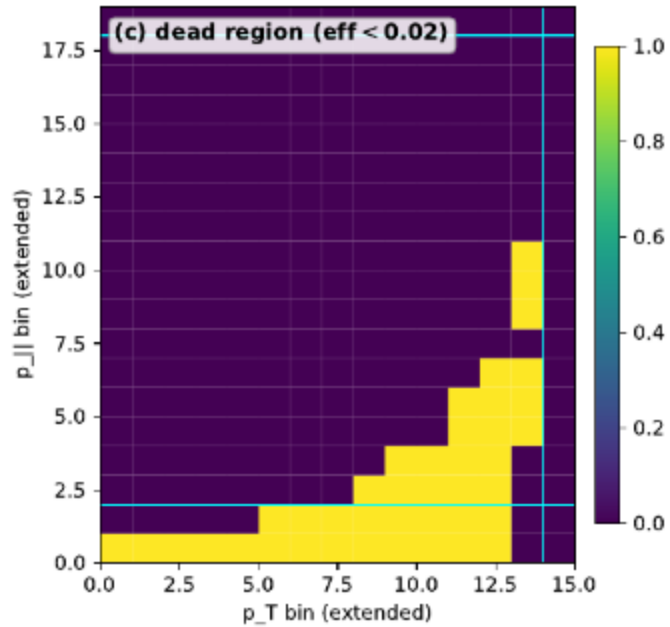
Regularization and detector response limit interpretation

Descriptive central values under the nominal response. The map does not establish a localized significance

Extending the domain increases reliance on simulation







Remove truth-muon kinematic cuts

27.7% of truth rate lies where efficiency < 2%

Keep the reconstructed MINOS selection

Central-value and prior tests only

Extended fiducial domain. No corrected covariance for this extension

The adopted 5D covariance carries four measurements

1. Seed sensitivity	6.145% projected-width movement versus a 5% bound
2. Tested ensemble sizes	6.02%, 6.02%, 6.145% at N = 40, 80, 160. One seed pair
3. Unprobed bands	Five seed-pinned bands: 6.75% of trace. Direction of missing effect unknown
4. Central shifts	Up to 6.02% of projected uncertainty. Up to 49.8% in individual 5D bins

Digest 3d7465f6… adopted under a byte-specific exception. Product fields remain NON-PASSING / adoptable: false

The scalar campaigns exposed two distinct biases

Background treatment	Purity subtraction biases the highest-W column by about 4% in nominal toys
Truth departures	Generator-sized deformations leave residuals up to 74% in tested cells
Targeted repair	More refinement-classifier capacity repairs nominal background bias
Remaining failure	Candidate R fails real-data numerical stability. No estimator adoption

A calibrated model test answers a narrower question

01

Assume a generator truth

Specify the fine-grid cross section and sub-grid model

This defines the null hypothesis

02

Simulate repeated experiments

Apply the declared nuisance variations and the estimator

Measure its behavior under that null

03

Compare the observed statistic

Count simulated tests at least as extreme

Correct for the ten comparisons

s5p measurement branch: NOT ADMITTED. The separate inference branch supplies no new reportable uncertainty

All ten tests reject their specified hybrid nulls

Five predictions × total and shape, familywise α = 0.05

1. Sub-grid shape is model-imposed. Refinement has not converged
2. Declared detector bands only. No recoil-energy-scale band
3. 34 interaction-model bands. Two negative-weight variations clipped
4. Half-MC toys versus full-MC data fit. Shift assessed and included
5. GiBUU has no Ev > 20 GeV. All predictions stop at 100 GeV
6. Generator finite-sample uncertainty enters the metric and draws

Conditional inference with candidate R, not a measured cross section. Independent recomputation agrees on the calculation

The inference has useful, measured limits

Grid sensitivity	All decisions remain determinate to $\kappa \approx 5.25$ along one tested shift direction
Response sensitivity	Exploratory ±4% data-side scale does not remove the rejections
Dimensional comparison	No qualified claim of extra power beyond matched coarse projections
Unresolved	The sub-grid residual and full hadronic-response uncertainty remain unmeasured

PET gives OmniFold a set of objects per event

Detector objects

Energy deposits in scintillator views



Truth particles

Four-momenta and particle identity



Muon event features condition the network

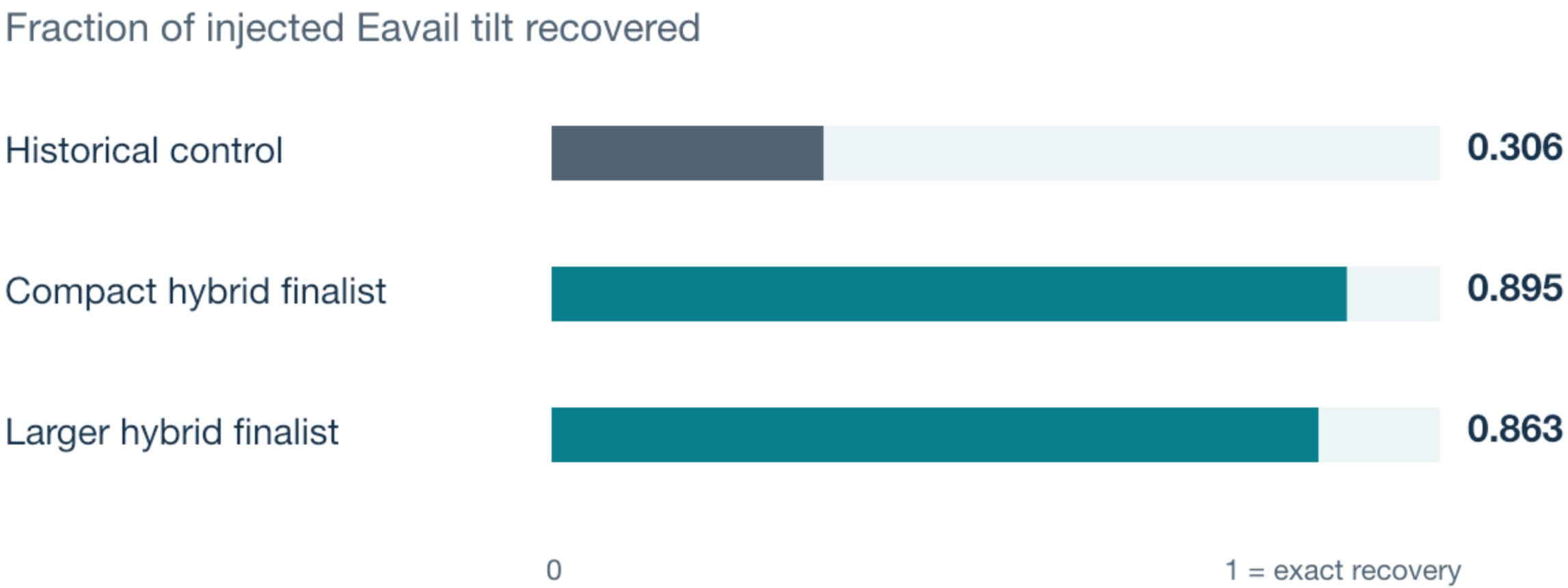
Attention learns object relationships; stored clouds retain at most 12 tokens

Historical recoil-only PET and the later full-event candidate have different input contracts

The earlier PET covariance did not qualify

Central/statistical pairing	Declined in August. A 50-member covariance exists but remains unverified
Observed anomaly	Nominal result lies outside its bootstrap family. Mechanism unresolved
Training variation	No full-event machine-learning covariance constructed. Gate 6 blocked
Scientific scope	Diagnostic and method development throughout

PET recovery improved substantially in simulation



Final-bank Eavail tilt, 60 draws. R = 1 is exact recovery, R = 0 retains the prior discrepancy

PET intervals fail differently across acceptance regions

Region	Nominal 68%	Nominal 95%
Aggregate Eavail	89.6%	99.0%
Low acceptance	**25.7%**	**78.9%**
Moderate acceptance	70.1%	96.4%
Good acceptance	92.9%	99.8%

Too wide in aggregate, too narrow at low acceptance

Compact finalist, 120 simulated draws, six bootstrap members per estimate. No eligible design or selected default

The paired PET–tree comparison depends on the distortion

Available-energy tilt	PET gains +0.102 and +0.070 in recovery over the matched tree comparator
Generator reweightings	Trees recover the three tested generator departures better
Hadron content	Some PET gains depend on the tree iteration choice
Scope	Exploratory point estimates on the same events. No uncertainty comparison

Reproducibility became part of the research

Execution integrity	An import-path defect could run another checkout's code despite matching file hashes
Scientific verification	Independent recomputation checks the calculation. It cannot validate every assumption
Stopping rules	Closed failed studies preserve useful limits and prevent retrospective selection
Deliverables	Note, primer, article and replay materials now separate claims by scope

Discussion with Ben

Article	Package assembled for review. Nothing submitted, deposited or sent
Measurement priority	Resolve the matched 2D estimator and uncertainty question before new production
Method development	PET recovery and interval calibration are separate next decisions
Scope decision	Which result should lead, and which validation is most valuable next?

Backup: exact available-energy convention

Truth	Protons: E – 938.27 MeV. Charged pions: E – 135 MeV
Truth continued	Neutral pions and photons: total energy. All other species contribute zero
Reconstruction	1.17 × (tracker + ECAL blob energy after muon-fuzz subtraction)
Comparison limit	The inherited charged-pion constant and species exclusions differ from some published definitions

Backup: q3 and W are inferred kinematic quantities

q0	Energy transfer: neutrino energy minus muon energy at truth level
Q^2	Positive spacelike four-momentum-transfer variable
q3	Magnitude of momentum transfer: $q3^2 = Q^2 + q0^2$
W	Experimenter's mass: $W^2 = M^2 + 2M\ q0 - Q^2$

Backup: the adopted covariance's exact boundary

Identity	z-cv.npz, SHA-256 prefix 3d7465f6, 890,500,272 bytes
Four measurements	M1–M4 on slide 13 travel with every use of this product
Unresolved criterion	Cause 3 remains predeclared and not computed for this digest
Decision meaning	Exception covers these bytes only. No inherited adoption for other candidates

Backup: reading the conditional p-values

Finite simulation	$p = (k + 1) / (B + 1)$, with k exceedances in B toys
Resolution	B = 1,343–1,751. Never report p = 0 or turn it into a Gaussian significance
Test domain	109 supported joint cells. 72 for GiBUU with $p_{\parallel} < 6$ GeV/c
Missing draws	224 calibration + 53 power draws recovered. Frozen decisions unchanged

Backup: the PET representation studies

Pooling versus individual objects	Synthetic routing comparison found no resolved accuracy difference
Production cloud	The top-12 cap removes substantial reconstructed activity
Improvement campaign	No frozen ten-iteration candidate passed adequacy plus stress
Final design and pretraining	No tested PET2 recipe passed every development screen

Backup: the closed scalar sequence

s5c	Futility stop. Nominal purity-background bias confirmed
s5n	Refined subtraction improves nominal behavior. Departure failure persists
s5e	Diagnosed refinement underfitting. Candidate R fails fresh assessment
s5p	Measurement not admitted. Conditional inference completes with no new uncertainty

Backup: what the release can reproduce

Available	Article sources, analysis note, primer and saved inference inputs
Replay	Recompute conditional inference from sufficient products
Different task	Regenerate training and calibration pseudo-experiments end to end
Release limits	Several closure, uncertainty and sensitivity results remain outside RC4 replay